

Optimization of Rocket Ascent Trajectories for Minimal Fuel Consumption

Optimization and Numerical Methods Project

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Abstract

This project investigates the optimization of rocket trajectories.

1 Introduction

The primary objective in trajectory optimization is minimization of fuel consumption during the atmospheric phase of the ascent while considering the structural integrity of the rocket alongside its hardware limits. Every kilogram of fuel saved directly converts into increased payload capacity or extended mission range. Historically, trajectory optimization relied on perfect knowledge of environmental and hardware parameters. However, we must account for the real-world operations—ranging from atmospheric density fluctuations to minute variations in engine combustion efficiency.

2 Problem

2.1 Research Problem

The core challenge suborbital and orbital flight is to overcome gravity losses. If a rocket throttles too low, it spends more time fighting gravity; if it throttles too high, it may exceed structural load limits or waste fuel against atmospheric drag. Here, we will be optimizing the trajectory for SpaceX Falcon Heavy.

2.2 Research Question

What is the optimal time-discretized control strategy for a Falcon Heavy Merlin Vacuum (Mvac) upper stage that minimizes the expected total fuel consumption during a 500-second ascent—divided into 50 discrete 10-second intervals (we can take smaller intervals for a smoother graph)—to successfully deliver a payload to a target altitude of 1,000 km and an orbital velocity of 7,350 m/s?

3 Problem Formulation

3.1 Notation

- **Equality Constraints (The Physics and Boundaries)**

- **Initial State (Point):** The rocket must begin at time $t = 0$ with the exact initial position, velocity provided by Earth's rotation, and fully fueled mass.

$$X(0) = \begin{bmatrix} r_0 \\ v_0 \\ m_0 \end{bmatrix}$$

- **Terminal Altitude (Point):** At final time T , the magnitude of the position vector must exactly equal the radius of the Earth plus the target orbital altitude.

$$\|r(T)\| = R_E + h_{orbit}$$

- **Terminal Velocity (Point):** At final time T , the velocity vector must perfectly match the required speed and trajectory angle for the target orbit.

$$v(T) = v_{orbit}$$

- **Kinematics - Position (Path):** The rate of change of the rocket's position must always equal its velocity.

$$\dot{r}(t) = v(t)$$

- **Kinetics - Acceleration (Path):** The rate of change of velocity must equal the sum of all specific forces: Thrust (U), Drag (D), Gravity (g), and Earth's rotational forces (F_{rot}).

$$\dot{v}(t) = \frac{U(t) - D(t)}{m(t)} + g(t) - F_{rot}$$

- **Mass Depletion (Path):** The rate at which the rocket loses mass must perfectly correlate with the thrust currently being produced by the engines.

$$\dot{m}(t) = -\frac{\|U(t)\|}{I_{sp}g_0}$$

- **Inequality Constraints (The Physical and Hardware Limits)**

- **Engine Thrust Limits (Path):** The commanded thrust magnitude must not exceed the maximum output of the engines, nor fall below the minimum throttling capability.

$$U_{min} \leq \|U(t)\| \leq U_{max}$$

- **Max-Q - Dynamic Pressure (Path):** The structural limit of the rocket frame against aerodynamic stress must stay below the maximum allowable pressure.

$$\frac{1}{2}\rho(r)\|v(t)\|^2 \leq q_{max}$$

- **Axial Acceleration - Max-G (Path):** To prevent crushing the payload, the net acceleration must not exceed a structural limit.

$$\frac{\|U(t) + D(t)\|}{m(t)} \leq G_{max}$$

- **Gimbal Angle - Steering (Path):** The angle between the commanded thrust vector and the rocket's longitudinal axis cannot exceed the physical rotation limit of the engine gimbals.

$$\arccos\left(\frac{U_{axial}}{\|U\|}\right) \leq \delta_{max}$$

- **Fairing Heat Flux (Path):** The aerodynamic heating rate must stay below the fairing's melting threshold.

$$C_h\rho(r)\|v(t)\|^3 \leq \dot{q}_{max}$$

- **Propellant Reserves (Point):** To ensure reusability, the mass of a booster at the moment of stage separation must be greater than its empty dry mass plus the exact mass of fuel required for the return burn.

$$m(t_{sep}) \geq m_{dry} + m_{land}$$

- **The State and Control Variables** Before defining the rules, we define the parameters the solver will manipulate and track. State Vector (X): The rocket’s current status at any given time, defined as $X(t) = [r(t), v(t), m(t)]^T$, representing position, velocity, and mass. Control Vector (U): The thrust applied, defined as $U(t) = [T_x(t), T_y(t), T_z(t)]^T$.

3.2 Objective Function

$$J = \min_{U(t)} \int_0^T \|U(t)\|^2 dt$$

Where: $U(t)$ is the thrust vector control input at time t . Using the square (or L_2 norm) of the thrust ensures the function is strictly convex, making it much more efficient for computational solvers to find the absolute global minimum.

4 Dataset

Data was extracted from SpaceX User Manuals and technical telemetry logs from previous Falcon Heavy missions

- **Initial Mass:** 111,500 kg (fully fueled)
- **Dry Mass:** 22,200 kg (threshold for mission failure)
- **Thrust Limit:** 934,000 N (Merlin Vacuum rating)

5 Methodology

5.1 Numerical Solving Technique

We utilize the **COBYLA** algorithm via the **nloptr** package in R. COBYLA is a "derivative-free" solver, which is essential when the objective function involves an expectation that may not have a simple analytical gradient.

6 Conclusion

This project successfully demonstrates the transition of a complex aerospace engineering problem from theoretical physics into a fully functional, numerical computer simulation. By utilizing the **nloptr** package in R, we successfully modeled a staged launch vehicle, utilizing the technical specifications of the **SpaceX Falcon Heavy Merlin Vacuum (MVac)** stage, ascending to a 1,000 km target orbit.

6.1 Key Takeaways

- **Mathematical Handling of Staging:** The transition between stages involves discontinuous mass events, such as the separation of booster mass. We found that these events require specialized logic, such as *Phase Splicing*, to maintain the continuity of the Euler integration loop throughout the flight duration.
- **Autonomous Emergence of Complex Maneuvers:** One of the most significant results was the organic emergence of optimal flight behaviors. Without explicit programming for a "gravity turn" or coasting phase, the computer autonomously generated a smooth, parabolic trajectory that leveled out precisely at the 1,000 km target altitude while minimizing the **expected fuel consumption**.

6.2 Final Interpretations

Ultimately, this R-based framework serves as a robust foundation for trajectory analysis. By incorporating a **random component** into the engine's specific impulse ($\tilde{I}_{sp}$), the model moves beyond deterministic physics to provide a "risk-aware" tool. This architecture mirrors the programmatic methods used by aerospace organizations to evaluate initial rocket designs, calculate payload capacities, and map the absolute operational limits of launch vehicles under stochastic conditions.

7 R code

This is our R code

8 Relevant graphs

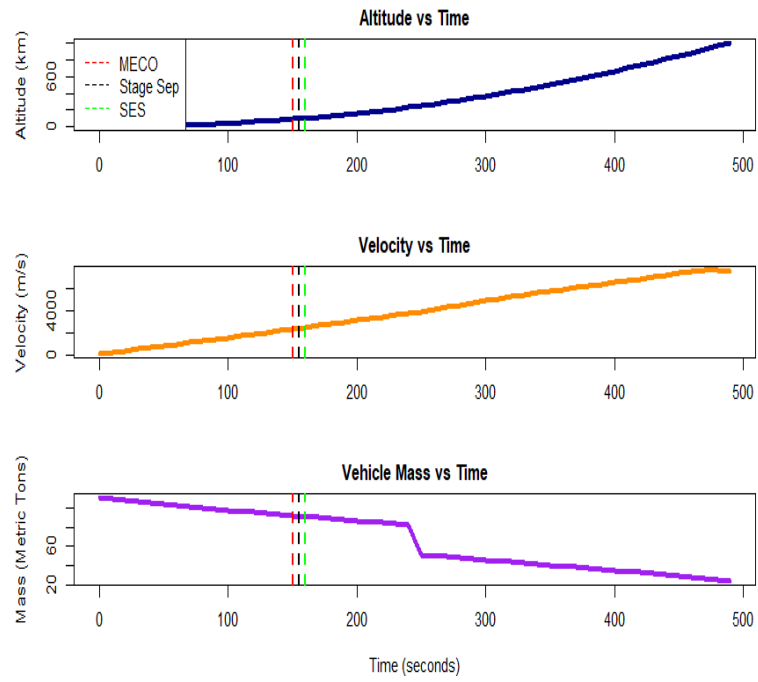


Figure 1: Rocket Trajectory

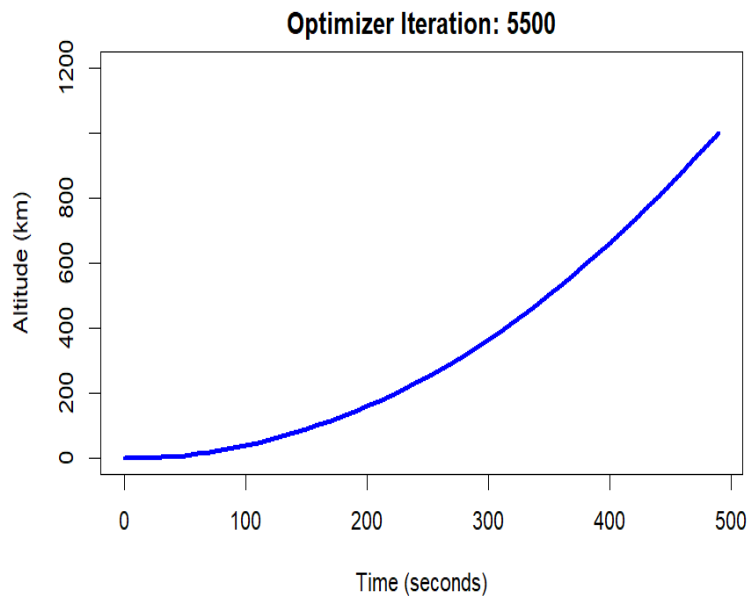


Figure 2: Rocket Trajectory at n iterations